

# Sealed Experiment Record

Sealed 2026-10-08 · Signed by the K-Veritas server  
Session 24aa1d59 · Data hash 02ac...5a9e

## Abstract

This record seals 1 run (kv\_run.sh) executed on Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz between 2026-10-08 03:45 and 06:13 UTC. Reported results: max\_senders\_per\_receiver 57, last\_flow\_finish\_us 135067. 1 invocation in total, 0 failed. Produced with K-Veritas [1].

### What this record proves

These numbers came from the sealed code at this time, unchanged since. Whether the experiment is sound is for the reader to judge.

Verify: kveritas verify <this file> or kveritas.org/verify

## 1 Setup

| Item               | Value                                                            |
|--------------------|------------------------------------------------------------------|
| Session            | 24aa1d59-b447-4308-a510-d747dafb5670                             |
| Initialized        | 2026-10-08T03:45:35Z                                             |
| Machine            | 167ec0d6dc9a8170                                                 |
| Operating system   | linux / amd64                                                    |
| CPU                | Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores                  |
| Memory             | 31.0 GB                                                          |
| GPU                | NVIDIA GeForce RTX 5060 Ti, 16311 MiB                            |
| Environment, run 1 | 2d079d310a3777c78784d14a6d98ea3b079170f2dcfb96c5205284e683ebb9e8 |
| Disclosure         | open                                                             |
| Source code, run 1 | 830dda188fdcf6f669d5972f9c3b4e1f41abe2c6a59817ea32b29ad47e472218 |

## 2 Runs

| # | Command        | Duration           | Exit | Anchored   |
|---|----------------|--------------------|------|------------|
| 1 | bash kv_run.sh | 2 hours 27 minutes | 0    | at run end |

1 invocation in total: 1 kept in this record, 0 failed, 0 interrupted. Every invocation is recorded by the server, including the ones not kept.

## 3 Results

### Run 1: kv\_run.sh

| Metric                | Final value | Step | Source   |
|-----------------------|-------------|------|----------|
| flows_finished        | 261632      |      | explicit |
| packets_new           | 6.5408e+07  |      | explicit |
| packets_retransmitted | 1.74483e+06 |      | explicit |

| Inline claim             | Value  | Phase | Output line |
|--------------------------|--------|-------|-------------|
| max_senders_per_receiver | 57     |       | 13          |
| last_flow_finish_us      | 135067 |       | 14          |

Seed committed: seed:13, output line 1, 2026-10-08T03:45:35Z.

## 4 Execution evidence

### Coherence

Coherent (0.99): the telemetry channels move together, as one process.

### Telemetry

| Run | Samples | CPU time | Peak memory | Peak GPU |
|-----|---------|----------|-------------|----------|
| 1   | 7200    | 8850.8 s | 15.21 GB    | 0%       |

## 5 Provenance

### Run 1: kv\_run.sh

Disclosure open. 290 tracked files, 3 snapshots. Every snapshot is bound into the signature.

| # | Snapshot  | Time     | Root             | Changed |
|---|-----------|----------|------------------|---------|
| 0 | run_start | 03:45:35 | 892b939bcfb88f53 | 290     |
| 2 | run_end   | 06:13:11 | 892b939bcfb88f53 | 0       |

1 snapshot with no file change omitted here; full timeline in Appendix B.

Activity: 291 files read, 6 files written, 7 subprocesses. Full list in Appendix B.

## 6 Verification

| Field            | Value                                                            |
|------------------|------------------------------------------------------------------|
| Signer           | Signed by the K-Veritas server                                   |
| Signed at        | 2026-10-08T06:13:13.750949+00:00                                 |
| Algorithm        | RSA-PSS-SHA256, 4096-bit key                                     |
| Key fingerprint  | eeac81653cf44ac029c4069cf2a06d9f                                 |
| Data hash        | 02accad8d67271c8c02c4dc3231ac875bc0f56a60f58cbd1b85ef7db7b1f5a9e |
| Code bundle hash | 44585e956d55b8e27b5baeae8fd83f2a07f33683c3c295fa17b2125dd2371c12 |

Run anchors: 1 invocations were anchored at the server when they ended.

### Checks a verifier runs

1. The stored canonical data hashes to the data hash above, and the signature verifies over data\_hash:nonce:signed\_at.
2. The signing key is the K-Veritas server key, not one supplied by the author.
3. Nothing was appended after the seal, and these pages hash to the value bound in the seal.
4. Each run matches the digest the server received when it ended.
5. A supplied code bundle hashes to the code bundle hash above.

## References

[1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.

# Appendix

## A Full metric series

### Run 1: kv\_run.sh

| Metric                | Value       | Step | Line | Source   |
|-----------------------|-------------|------|------|----------|
| flows_finished        | 261632      |      | 10   | explicit |
| packets_new           | 6.5408e+07  |      | 11   | explicit |
| packets_retransmitted | 1.74483e+06 |      | 12   | explicit |

| Phase    | Line | Time     | CPU time | Memory  | GPU |
|----------|------|----------|----------|---------|-----|
| simulate | 2    | 03:45:36 | 0.5 s    | 0.02 GB |     |

## B Provenance timeline and activity map

### Run 1: kv\_run.sh

| # | Snapshot       | Time     | Root             | Changed |
|---|----------------|----------|------------------|---------|
| 0 | run_start      | 03:45:35 | 892b939bcfb88f53 | 290     |
| 1 | phase simulate | 03:45:36 | 892b939bcfb88f53 | 0       |
| 2 | run_end        | 06:13:11 | 892b939bcfb88f53 | 0       |

| Event | Path or command                                                              | Hash |
|-------|------------------------------------------------------------------------------|------|
| read  | kv_run.sh                                                                    |      |
| read  | experiments/fattree-all-to-all/sequential_512conns/run.sh                    |      |
| read  | sim/datacenter/connection_matrices/gen_serialn_alltoall.py                   |      |
| read  | experiments/fattree-all-to-all/sequential_512conns/plot_cdf.py               |      |
| read  | experiments/fattree-all-to-all/sequential_512conns/plot_overlap.py           |      |
| read  | experiments/fattree-all-to-all/sequential_512conns/proc_incast_color.py      |      |
| read  | experiments/fattree-all-to-all/sequential_512conns_prio/plot_cdf.py          |      |
| read  | experiments/fattree-all-to-all/sequential_512conns_prio/plot_overlap.py      |      |
| read  | experiments/fattree-all-to-all/sequential_512conns_prio/proc_incast_color.py |      |
| read  | experiments/fattree-all-to-all/sequential_512conns_prio/run.sh               |      |
| read  | experiments/fattree-spraying/oblivious_trim_ecn/3tier_notrim_256/run.sh      |      |
| read  | experiments/fattree-spraying/oblivious_trim_ecn/3tier_notrim_32/run.sh       |      |
| read  | experiments/fattree-spraying/oblivious_trim_ecn/3tier_notrim_ecn_256/run.sh  |      |
| read  | experiments/fattree-spraying/oblivious_trim_ecn/3tier_notrim_ecn_32/run.sh   |      |
| read  | experiments/fattree-spraying/oblivious_trim_ecn/3tier_trim_256/run.sh        |      |
| read  | experiments/fattree-spraying/oblivious_trim_ecn/3tier_trim_32/run.sh         |      |
| read  | experiments/fattree-spraying/oblivious_trim_ecn/run.sh                       |      |
| read  | experiments/fattree-spraying/oblivious_trim_ecn/run_sequential.sh            |      |
| read  | experiments/fattree-spraying/spray_comparison/makegraph.py                   |      |
| read  | experiments/fattree-spraying/spray_comparison/run.sh                         |      |
| read  | sim/EXAMPLES/collateral/process_dctcp_collateral.py                          |      |
| read  | sim/EXAMPLES/collateral/process_dctcp_collateral_lossless.py                 |      |
| read  | sim/EXAMPLES/collateral/process_ndp_collateral.py                            |      |
| read  | sim/EXAMPLES/collateral/run.sh                                               |      |
| read  | sim/EXAMPLES/collateral/run_dctcp_collateral.sh                              |      |
| read  | sim/EXAMPLES/collateral/run_dctcp_collateral_lossless.sh                     |      |
| read  | sim/EXAMPLES/collateral/run_ndp_collateral.sh                                |      |

| Event | Path or command                                                 | Hash |
|-------|-----------------------------------------------------------------|------|
| read  | sim/EXAMPLES/failure/process_data.py                            |      |
| read  | sim/EXAMPLES/failure/process_dctcp_data.py                      |      |
| read  | sim/EXAMPLES/failure/process_mptcp_data.py                      |      |
| read  | sim/EXAMPLES/failure/run.sh                                     |      |
| read  | sim/EXAMPLES/in_and_out/process_data.py                         |      |
| read  | sim/EXAMPLES/in_and_out/run.sh                                  |      |
| read  | sim/EXAMPLES/incast_scaling/process_data_incast_conns.py        |      |
| read  | sim/EXAMPLES/incast_scaling/run.sh                              |      |
| read  | sim/EXAMPLES/incast_shortflows/process_data_incast_conns.py     |      |
| read  | sim/EXAMPLES/incast_shortflows/run.sh                           |      |
| read  | sim/EXAMPLES/permutation/process_data.py                        |      |
| read  | sim/EXAMPLES/permutation/process_dctcp_data.py                  |      |
| read  | sim/EXAMPLES/permutation/process_mptcp_data.py                  |      |
| read  | sim/EXAMPLES/permutation/run.sh                                 |      |
| read  | sim/aeolusqueue.cpp                                             |      |
| read  | sim/aeolusqueue.h                                               |      |
| read  | sim/callback_pipe.cpp                                           |      |
| read  | sim/callback_pipe.h                                             |      |
| read  | sim/cbr.cpp                                                     |      |
| read  | sim/cbr.h                                                       |      |
| read  | sim/cbrpacket.cpp                                               |      |
| read  | sim/cbrpacket.h                                                 |      |
| read  | sim/circular_buffer.h                                           |      |
| read  | sim/clock.cpp                                                   |      |
| read  | sim/clock.h                                                     |      |
| read  | sim/compositeprioqueue.cpp                                      |      |
| read  | sim/compositeprioqueue.h                                        |      |
| read  | sim/compositequeue.cpp                                          |      |
| read  | sim/compositequeue.h                                            |      |
| read  | sim/config.cpp                                                  |      |
| read  | sim/config.h                                                    |      |
| read  | sim/cpqueue.cpp                                                 |      |
| read  | sim/cpqueue.h                                                   |      |
| read  | sim/datacenter/bcube_topology.cpp                               |      |
| read  | sim/datacenter/bcube_topology.h                                 |      |
| read  | sim/datacenter/camcubetopology.cpp                              |      |
| read  | sim/datacenter/camcubetopology.h                                |      |
| read  | sim/datacenter/connection_matrices/gen_allreduce.py             |      |
| read  | sim/datacenter/connection_matrices/gen_allreduce_butterfly.py   |      |
| read  | sim/datacenter/connection_matrices/gen_incast.py                |      |
| read  | sim/datacenter/connection_matrices/gen_outcast_incast.py        |      |
| read  | sim/datacenter/connection_matrices/gen_permutation.py           |      |
| read  | sim/datacenter/connection_matrices/gen_serial_alltoall.py       |      |
| read  | sim/datacenter/connection_matrices/gen_serialn_alltoall_prio.py |      |
| read  | sim/datacenter/connection_matrix.cpp                            |      |
| read  | sim/datacenter/connection_matrix.h                              |      |
| read  | sim/datacenter/dragon_fly_topology.cpp                          |      |
| read  | sim/datacenter/dragon_fly_topology.h                            |      |
| read  | sim/datacenter/fat_tree_switch.cpp                              |      |
| read  | sim/datacenter/fat_tree_switch.h                                |      |

| Event | Path or command                                                    | Hash |
|-------|--------------------------------------------------------------------|------|
| read  | sim/datacenter/fat_tree_topology.cpp                               |      |
| read  | sim/datacenter/fat_tree_topology.h                                 |      |
| read  | sim/datacenter/firstfit.cpp                                        |      |
| read  | sim/datacenter/firstfit.h                                          |      |
| read  | sim/datacenter/generic_topology.cpp                                |      |
| read  | sim/datacenter/generic_topology.h                                  |      |
| read  | sim/datacenter/incast.cpp                                          |      |
| read  | sim/datacenter/incast.h                                            |      |
| read  | sim/datacenter/main.h                                              |      |
| read  | sim/datacenter/main_eqds.cpp                                       |      |
| read  | sim/datacenter/main_hpcc.cpp                                       |      |
| read  | sim/datacenter/main_ndp.cpp                                        |      |
| read  | sim/datacenter/main_roce.cpp                                       |      |
| read  | sim/datacenter/main_swift.cpp                                      |      |
| read  | sim/datacenter/main_tcp.cpp                                        |      |
| read  | sim/datacenter/mtcp_subflow_control.cpp                            |      |
| read  | sim/datacenter/mtcp_subflow_control.h                              |      |
| read  | sim/datacenter/multihomed_fat_tree_topology.cpp                    |      |
| read  | sim/datacenter/multihomed_fat_tree_topology.h                      |      |
| read  | sim/datacenter/old_tests/main_dctcp_incast_collateral.cpp          |      |
| read  | sim/datacenter/old_tests/main_dctcp_incast_collateral_lossless.cpp |      |
| read  | sim/datacenter/old_tests/main_dctcp_permutation.cpp                |      |
| read  | sim/datacenter/old_tests/main_hpcc.cpp                             |      |
| read  | sim/datacenter/old_tests/main_ndp2.cpp                             |      |
| read  | sim/datacenter/old_tests/main_ndp_alltoall.cpp                     |      |
| read  | sim/datacenter/old_tests/main_ndp_cmcube_permutation.cpp           |      |
| read  | sim/datacenter/old_tests/main_ndp_dragonfly_permutation.cpp        |      |
| read  | sim/datacenter/old_tests/main_ndp_in_out.cpp                       |      |
| read  | sim/datacenter/old_tests/main_ndp_incast.cpp                       |      |
| read  | sim/datacenter/old_tests/main_ndp_incast2.cpp                      |      |
| read  | sim/datacenter/old_tests/main_ndp_incast2b.cpp                     |      |
| read  | sim/datacenter/old_tests/main_ndp_incast_collateral.cpp            |      |
| read  | sim/datacenter/old_tests/main_ndp_incast_shortflows.cpp            |      |
| read  | sim/datacenter/old_tests/main_ndp_incast_tor.cpp                   |      |
| read  | sim/datacenter/old_tests/main_ndp_outcast.cpp                      |      |
| read  | sim/datacenter/old_tests/main_ndp_outcast_incast.cpp               |      |
| read  | sim/datacenter/old_tests/main_ndp_outcast_shortflows.cpp           |      |
| read  | sim/datacenter/old_tests/main_ndp_oversubscribed.cpp               |      |
| read  | sim/datacenter/old_tests/main_ndp_permutation.cpp                  |      |
| read  | sim/datacenter/old_tests/main_ndp_permutation_fail.cpp             |      |
| read  | sim/datacenter/old_tests/main_ndp_permutation_incast.cpp           |      |
| read  | sim/datacenter/old_tests/main_ndp_permutation_lossless.cpp         |      |
| read  | sim/datacenter/old_tests/main_ndp_permutation_tor.cpp              |      |
| read  | sim/datacenter/old_tests/main_ndp_random.cpp                       |      |
| read  | sim/datacenter/old_tests/main_ndp_random_shortflows_tor.cpp        |      |
| read  | sim/datacenter/old_tests/main_ndprts_alltoall.cpp                  |      |
| read  | sim/datacenter/old_tests/main_ndprts_outcast.cpp                   |      |
| read  | sim/datacenter/old_tests/main_ndprts_outcast_incast.cpp            |      |
| read  | sim/datacenter/old_tests/main_ndprts_permutation_incast.cpp        |      |
| read  | sim/datacenter/old_tests/main_ndptunnel_incast_shortflows.cpp      |      |

| Event | Path or command                                             | Hash |
|-------|-------------------------------------------------------------|------|
| read  | sim/datacenter/old_tests/main_ndptunnel_permutation.cpp     |      |
| read  | sim/datacenter/old_tests/main_swift_incast.cpp              |      |
| read  | sim/datacenter/old_tests/main_swift_incast2.cpp             |      |
| read  | sim/datacenter/old_tests/main_swift_incast2b.cpp            |      |
| read  | sim/datacenter/old_tests/main_swift_incast3.cpp             |      |
| read  | sim/datacenter/old_tests/main_tcp_dragonfly_permutation.cpp |      |
| read  | sim/datacenter/old_tests/main_tcp_outcast_incast.cpp        |      |
| read  | sim/datacenter/old_tests/main_tcp_perm_shortflows.cpp       |      |
| read  | sim/datacenter/old_tests/main_tcp_permutation.cpp           |      |
| read  | sim/datacenter/oversubscribed_fat_tree_topology.cpp         |      |
| read  | sim/datacenter/oversubscribed_fat_tree_topology.h           |      |
| read  | sim/datacenter/shortflows.cpp                               |      |
| read  | sim/datacenter/shortflows.h                                 |      |
| read  | sim/datacenter/star_topology.cpp                            |      |
| read  | sim/datacenter/star_topology.h                              |      |
| read  | sim/datacenter/subflow_control.cpp                          |      |
| read  | sim/datacenter/subflow_control.h                            |      |
| read  | sim/datacenter/topology.h                                   |      |
| read  | sim/datacenter/validate.py                                  |      |
| read  | sim/datacenter/vl2_topology.cpp                             |      |
| read  | sim/datacenter/vl2_topology.h                               |      |
| read  | sim/dctcp.cpp                                               |      |
| read  | sim/dctcp.h                                                 |      |
| read  | sim/dctcp_transfer.cpp                                      |      |
| read  | sim/dctcp_transfer.h                                        |      |
| read  | sim/drawable.h                                              |      |
| read  | sim/ecn.h                                                   |      |
| read  | sim/ecnprioqueue.cpp                                        |      |
| read  | sim/ecnprioqueue.h                                          |      |
| read  | sim/ecnqueue.cpp                                            |      |
| read  | sim/ecnqueue.h                                              |      |
| read  | sim/eqds.cpp                                                |      |
| read  | sim/eqds.h                                                  |      |
| read  | sim/eqds_logger.cpp                                         |      |
| read  | sim/eqds_logger.h                                           |      |
| read  | sim/eqdspacket.cpp                                          |      |
| read  | sim/eqdspacket.h                                            |      |
| read  | sim/eth_pause_packet.cpp                                    |      |
| read  | sim/eth_pause_packet.h                                      |      |
| read  | sim/eventlist.cpp                                           |      |
| read  | sim/eventlist.h                                             |      |
| read  | sim/exoqueue.cpp                                            |      |
| read  | sim/exoqueue.h                                              |      |
| read  | sim/faircompositequeue.cpp                                  |      |
| read  | sim/faircompositequeue.h                                    |      |
| read  | sim/fairpullqueue.cpp                                       |      |
| read  | sim/fairpullqueue.h                                         |      |
| read  | sim/hpcc.cpp                                                |      |
| read  | sim/hpcc.h                                                  |      |
| read  | sim/hpccpacket.cpp                                          |      |

| Event | Path or command               | Hash |
|-------|-------------------------------|------|
| read  | sim/hpccpacket.h              |      |
| read  | sim/logfile.cpp               |      |
| read  | sim/logfile.h                 |      |
| read  | sim/loggers.cpp               |      |
| read  | sim/loggers.h                 |      |
| read  | sim/loggertypes.h             |      |
| read  | sim/matrix.h                  |      |
| read  | sim/meter.cpp                 |      |
| read  | sim/meter.h                   |      |
| read  | sim/mpswift.cpp               |      |
| read  | sim/mpswift.h                 |      |
| read  | sim/mtcp.cpp                  |      |
| read  | sim/mtcp.h                    |      |
| read  | sim/ndp-orig.cpp              |      |
| read  | sim/ndp.cpp                   |      |
| read  | sim/ndp.h                     |      |
| read  | sim/ndp_transfer.cpp          |      |
| read  | sim/ndp_transfer.h            |      |
| read  | sim/ndppacket.cpp             |      |
| read  | sim/ndppacket.h               |      |
| read  | sim/ndptunnel.cpp             |      |
| read  | sim/ndptunnel.h               |      |
| read  | sim/ndptunnelpacket.cpp       |      |
| read  | sim/ndptunnelpacket.h         |      |
| read  | sim/network.cpp               |      |
| read  | sim/network.h                 |      |
| read  | sim/parse_output.cpp          |      |
| read  | sim/pipe.cpp                  |      |
| read  | sim/pipe.h                    |      |
| read  | sim/priopullqueue.cpp         |      |
| read  | sim/priopullqueue.h           |      |
| read  | sim/prioqueue.cpp             |      |
| read  | sim/prioqueue.h               |      |
| read  | sim/qcn.cpp                   |      |
| read  | sim/qcn.h                     |      |
| read  | sim/queue.cpp                 |      |
| read  | sim/queue.h                   |      |
| read  | sim/queue_lossless.cpp        |      |
| read  | sim/queue_lossless.h          |      |
| read  | sim/queue_lossless_input.cpp  |      |
| read  | sim/queue_lossless_input.h    |      |
| read  | sim/queue_lossless_output.cpp |      |
| read  | sim/queue_lossless_output.h   |      |
| read  | sim/randomqueue.cpp           |      |
| read  | sim/randomqueue.h             |      |
| read  | sim/rcppacket.cpp             |      |
| read  | sim/rng.cpp                   |      |
| read  | sim/roce.cpp                  |      |
| read  | sim/roce.h                    |      |
| read  | sim/rocepacket.cpp            |      |

| Event | Path or command                                          | Hash |
|-------|----------------------------------------------------------|------|
| read  | sim/rocepacket.h                                         |      |
| read  | sim/route.cpp                                            |      |
| read  | sim/route.h                                              |      |
| read  | sim/routetable.cpp                                       |      |
| read  | sim/routetable.h                                         |      |
| read  | sim/rtx_timer.cpp                                        |      |
| read  | sim/rtx_timer.h                                          |      |
| read  | sim/sent_packets.cpp                                     |      |
| read  | sim/sent_packets.h                                       |      |
| read  | sim/strack.cpp                                           |      |
| read  | sim/strack.h                                             |      |
| read  | sim/strackpacket.cpp                                     |      |
| read  | sim/strackpacket.h                                       |      |
| read  | sim/swift.cpp                                            |      |
| read  | sim/swift.h                                              |      |
| read  | sim/swift_new.cpp                                        |      |
| read  | sim/swift_new.h                                          |      |
| read  | sim/swift_scheduler.cpp                                  |      |
| read  | sim/swift_scheduler.h                                    |      |
| read  | sim/swift_transfer.cpp                                   |      |
| read  | sim/swift_transfer.h                                     |      |
| read  | sim/swiftpacket.cpp                                      |      |
| read  | sim/swiftpacket.h                                        |      |
| read  | sim/switch.cpp                                           |      |
| read  | sim/switch.h                                             |      |
| read  | sim/tcp.cpp                                              |      |
| read  | sim/tcp.h                                                |      |
| read  | sim/tcp_periodic.cpp                                     |      |
| read  | sim/tcp_periodic.h                                       |      |
| read  | sim/tcp_transfer.cpp                                     |      |
| read  | sim/tcp_transfer.h                                       |      |
| read  | sim/tcppacket.cpp                                        |      |
| read  | sim/tcppacket.h                                          |      |
| read  | sim/tests/htsim-tests/bidir_ndp/test_config.json         |      |
| read  | sim/tests/htsim-tests/bidir_swift/test_config.json       |      |
| read  | sim/tests/htsim-tests/dumbell_hpcc/test_config.json      |      |
| read  | sim/tests/htsim-tests/dumbell_ndp/test_config.json       |      |
| read  | sim/tests/htsim-tests/dumbell_ndptunnel/test_config.json |      |
| read  | sim/tests/htsim-tests/dumbell_roce/test_config.json      |      |
| read  | sim/tests/htsim-tests/dumbell_strack/test_config.json    |      |
| read  | sim/tests/htsim-tests/dumbell_swift/test_config.json     |      |
| read  | sim/tests/htsim-tests/dumbell_tcp/test_config.json       |      |
| read  | sim/tests/htsim-tests/multihop_swift/test_config.json    |      |
| read  | sim/tests/htsim-tests/multihop_swift2/test_config.json   |      |
| read  | sim/tests/htsim-tests/multipath_swift/test_config.json   |      |
| read  | sim/tests/htsim-tests/trigger_test/test_config.json      |      |
| read  | sim/tests/main.cpp                                       |      |
| read  | sim/tests/main_bidir_ndp.cpp                             |      |
| read  | sim/tests/main_bidir_swift.cpp                           |      |
| read  | sim/tests/main_dumbell_hpcc.cpp                          |      |

| Event   | Path or command                                                                 | Hash             |
|---------|---------------------------------------------------------------------------------|------------------|
| read    | sim/tests/main_dumbell_ndp.cpp                                                  |                  |
| read    | sim/tests/main_dumbell_ndptunnel.cpp                                            |                  |
| read    | sim/tests/main_dumbell_roce.cpp                                                 |                  |
| read    | sim/tests/main_dumbell_strack.cpp                                               |                  |
| read    | sim/tests/main_dumbell_swift.cpp                                                |                  |
| read    | sim/tests/main_dumbell_tcp.cpp                                                  |                  |
| read    | sim/tests/main_multihop_swift.cpp                                               |                  |
| read    | sim/tests/main_multihop_swift2.cpp                                              |                  |
| read    | sim/tests/main_multipath_swift.cpp                                              |                  |
| read    | sim/tests/main_trigger_test.cpp                                                 |                  |
| read    | sim/tests/tests.py                                                              |                  |
| read    | sim/trigger.cpp                                                                 |                  |
| read    | sim/trigger.h                                                                   |                  |
| write   | experiments/fattree-all-to-all/sequential_512conns/a2a-1024-512-1.cm            | 5aa4828e735a5fcc |
| read    | sim/datacenter/htsim_ndp                                                        |                  |
| write   | experiments/fattree-all-to-all/sequential_512conns/idmap.txt                    | f1b469dc88e51f4c |
| write   | experiments/fattree-all-to-all/sequential_512conns/logout.dat                   | c3e358a545d93033 |
| write   | experiments/fattree-all-to-all/sequential_512conns/out_1024_512_perm_50iw...    | 4e7fcba9f644ca5f |
| write   | experiments/fattree-all-to-all/sequential_512conns/incast_1024_512_perm_50...   | 156a10b0a45999f1 |
| write   | experiments/fattree-all-to-all/sequential_512conns/incast_1024_512_perm_50...   | 3fe6131251708bf8 |
| process | bash run.sh                                                                     |                  |
| process | python ../../sim/datacenter/connection_matrices/gen_serialn_alltoall.py a2a-... |                  |
| process | grep -v gnuplot: command not found                                              |                  |
| process | ../../sim/datacenter/htsim_ndp -tm a2a-1024-512-1.cm -linkspeed 100000 -st...   |                  |
| process | python plot_overlap.py 1024 512 1 50                                            |                  |
| process | python proc_incant_color.py incant_1024_512_perm_50iw_1par.tmp 1                |                  |
| process | python -                                                                        |                  |

## C Run history

### Invocation 1, OK, 2 hours 27 minutes, 14 output lines, started 2026-10-08T03:45:35

metric\_hash 323752c9bb12211d1be7f6f4f81eb55f96819aeb8523c887097a397eb136b02a  
run\_digest a8671cb0fcef317990e030f7e874163b491fb3b9b4bf547585a15c6c0eb3668f  
anchored at run end

## D Raw seal fields and hashes

### Seal

data\_hash 02accad8d67271c8c02c4dc3231ac875bc0f56a60f58cbd1b85ef7db7b1f5a9e  
signed\_message\_hash 801858efe66f82202a24313c44d9b7c301e858b7f2514ac5e5e31577021e8a78  
nonce 5fe2c21444511e88e805d9be8a263cb7  
signed\_at 2026-10-08T06:13:13.750949+00:00  
sealed\_at 2026-10-08T06:13:13.817844821Z  
checkout\_bundle 44585e956d55b8e27b5baeae8fd83f2a07f33683c3c295fa17b2125dd2371c12

### Signature (base64)

X8uERaR7lHoY6qh2DqbQUwL046S5iu/QQ4u8S6xp0o/xihHSAydwqhiKGCWfuKbMVB56/Amzcw2Cbr/b99vp1ekz  
tP2PMLlWLxxTPy7RmPsI3rrcVNqgZ4o3tx0MYIwlnSiATlwtgzph1fAURtMPb2wj/ktXxG87jZPTax/u7a0Bizdb  
K/jjYN34bBuQ0CpJULo4pmLzQCe3ww0pK79yi33RT9AKYDPG6qGV3aVw2iWItL/rKK/7gmwNdSqIHPFMSp0LMiZe  
QUs0Q75++G528CEWJglogHwMvVxks+0b5xCDF77870E01w8xvNwo7i6yVEDde16brRuSCqkek6CCb85j5WrVdJNJ  
UqNhsKVi0Nq4BbmWBuY7TAr5i4llWxtxy3D8CHWENwPmIJj/DVH0oGt+sGdd2GfYsULbTVYizDn/GTDE5UbfkI  
wmBH048Y/v7/2YIA0ezvYiStvTw2L8hHWzx8KBfm39mwL3wlTh08pZMTS6+sowGjRYNAp8PcVTcaNEQF8JOYhQFJ

R+xYRLuynQMh6lPnCvJB/x5gj0Vn5nF/K+JneGa0MEzhoRmXVe5HzlF09cJqzUMCVnE01c/4+w5mIwdJPm6dEidU  
eabzGN2kFSK1tB7AUXDZCAknJo43Db89YmoZ9EGjo05hto54hPvB+2vcrFj1dPFEIHc=

## Public key

-----BEGIN PUBLIC KEY-----

MIICIjANBgkqhkiG9w0BAQEFAAACAg8AMIICCgKCAgEA1QL3YyMgeShx8IhTC7jb  
sM0w2BnRXgsBn3+SMXKjtwHATqdekmlPlcjayJxD5fcPrOPPCqp/LouFz5CUZzRi  
6+8FTv70dHuIz5A4Lat4vcgWP+iWscPc3HzpmjpsKbTm0emqgp6RB6GNNPpxYa5X  
7qU95EfEisX/GfoS9ToXtqPBwCcVWSPBX8efeC4TC58m96Fsp4mC8KIcEadbIbHN  
av0JjdTJ//CMOWZSIbdzE6qvEbNFCqDB5tNU7oHfYvBeNLrvPU9TAqdbtozdHQ+D  
AhfTRuJNgWEKjwD5lb0Vhi4qlkweSOEMTKT3ye9Il5xEaJn4bku4eKqeS7VSZt3o  
XbUcmr2H5mKcKSvt6bXmX6ajAmgi900Cm9QJlaQkJM6z2iITnyvDMgueJtFxQpC9  
HNBPgPZ242NH8JgE6CNF7ToL9syasS67u7UMgIcNgeNHNcy9bLgPU5annzP9gN2l  
6HEMLdkXNdKQRb1rDwyEJf3N/50YHnXXPmuf5sb8R/pxwNqtF04R5UGVV92qd2Mi  
1GN7lE3mbQRbPsLL/JApDXMNzNikaFfBsQK7iKxr7vIaTeZckFZWfThcYNT7Hasg  
+eE4bf/olCHBCf1VIDCKDxohehdZ0WRRcgM9iK3I/0zZdaBsgdc8WuJRmb8LPP9v  
8Jap/1PWjncNku5cdW3gHLCcAwEAAQ==

-----END PUBLIC KEY-----

## Run 1 outputs

```
stdout d9ba860b9a1dfe57e8e56d06148a7d94b2aca54e786ed9b783be43eb193cf647
stderr e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855
metrics 323752c9bb12211d1be7f6f4f81eb55f96819aeb8523c887097a397eb136b02a
digest a8671cb0fcef317990e030f7e874163b491fb3b9b4bf547585a15c6c0eb3668f
kv_run.sh (unchanged)
pre 55e3aa10520fd3d78880781c493c4e386561c57be28f4e4d792d067c05ac2e23
post 55e3aa10520fd3d78880781c493c4e386561c57be28f4e4d792d067c05ac2e23
```

## E BibTeX for [1]

```
@misc{keita2026computerscienceconferencesrequire,
  title={Computer Science Conferences Should Require Nonrepudiable Experimental Results},
  author={Mamadou K. Keita and Christopher Homan},
  year={2026},
  eprint={2605.08586},
  archivePrefix={arXiv},
  primaryClass={cs.CR},
  url={https://arxiv.org/abs/2605.08586},
}
```